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Quantification of depression disorder using EEG signal

Mojtaba Hajian
Biomedical Engineering department
Amirkabir University of Technology, AUT
Tehran, Iran
M_hajian@aut.ac.ir

Mohammad Hassan Moradi
Biomedical Engineering Department
Amirkabir University of Technology, AUT
Tehran, Iran
mhmoradi@aut.ac.ir

Abstract— **Depression is one of the most common psychological disorders, which has been an ever-growing concern in science world. In mental health questionnaires, including Beck's questionnaire, the patient is assigned a numerical indicator. Researches have shown that level of depression in people is associated with structural changes in the brain, therefore, by analyzing brain signals, it is possible to detect depression level. This paper presents a method that estimates the beck's index of each subject by extracting specific features from the patient's EEG signal. In order to quantify depression, an algorithm has been designed and implemented that uses membership values obtained from the fuzzy classifier and the support vector machine(SVM). In this approach, desirable results have been obtained indicating that the proposed algorithm has a good ability to determine the numerical index for depression. The results were obtained with a percent relative difference(PRD) of 5% and a Pearson correlation of 0.92. The results of the experiments show that the estimated numerical value of the designed system is of high correlation and low amount of PRD in comparison with the original beck number, related to each person.**

Keywords— ***Depression, EEG signal, fuzzy classifier, Beck's depression quastionnaire, quantification***

I. Introduction

According to the World Health Organization (WHO) anticipation, by 2020, after cardiovascular disease (CVD), depression is the second most significant disorder that puts a heavy burden on societies. Meanwhile, almost half of the patients are unaware of their disease, or their disorder have been diagnosed incorrectly. Therefore, accurate diagnosis of the disease is essential. The main method for diagnosis of depression is oral interview between the expert and the patient and written questionnaires that are designed based on standard criteria for diagnosis of the disorder. This type of depression diagnosis is not accurate, so using a brain physiology approach, hence, can be a valuable aid in detecting the level of depression. Many studies have been conducted to classify the level of depression. Depression as a brain and cognitive disorder shows a significant effect on the brain-related signals[1] specially the EEG signal[2]. Many studies have suggested that the level of depression can be achieved by processing the EEG signal of subjects. In[3], the features of the signals were extracted using linear methods such as statistical and frequency features, and also nonlinear methods such as chaos and the criterion of complexity(Lempel-Ziv) and correlation dimension. Linear Discriminant Analysis(LDA), Logistic Regression(LR) and KNN methods were applied for binary classification to discriminate depressed and healthy subjects. The best accuracy achieved by LR method with chaotic features was 92%. The LDA method and nonlinear features were used for multiclass classification of four depression levels (weak, mild, moderate, severe), with the best accuracy of 90%.

In another study, 33 depressed patients and 30 healthy subjects were separated. For this purpose, the power of different frequency bands of the EEG signal and the inter-hemispheric asymmetry of the alpha wave were selected as features, and the optimal features were distinguished based on ROC and then classified. More valuable features were chosen and given to the LR, SVM and Naïve Bayes(NB) classifiers. The reliable results were 97.6% for the LR method , 98.4% for SVM method and 96.8% for the NB method [4]. Those aforementioned researches show that there is a significant relationship between depression and EEG signal.

There have been so many studies to classify depression, and studies on the quantification of T-wave alternans[5] and estimation of depth of anesthesia[6, 7]. Almost nothing has been done for indexing depression and numerical assignments that is based on physiology. In [8], features were extracted using nonlinear analysis such as Largest Liapunov Exponent(LLE), entropy, Detrended Fluctuation Analysis(DFA), hurst's exponent, High Order of Spectra(HOS) and Recurrence Quantification Analysis(RQA), features were extracted and ranked by t-value method and the best features were put in linear relationships which gives the value of the numerical index of depression. This index is not precise and only has the ability to distinguish two classes (depressed-healthy).

In this research, our goal is to produce a continuous number that is supposed to show the exact level of depression. The criterion for this number is a numerical value derived from the Beck Depression Inventory[9], which is a numeric value between 1 and 63, which indicates the level of the depression severity.

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The rest of the paper is organized as follows: In section II, data acquisition and the algorithm have been described and, experimental results are indicated in section III. In section IV the conclusion of the paper is explained.

II. MATERIAL AND METHODS

A. Data Acquisition

In this study, EEG data of 75 depressed patients were used with 4 levels of depression (22 cases of weak, 11 cases of mild, 11 cases of moderate, and 35 cases of severe levels of depression). Detection of the presence or absence of depression was based on the opinion of psychiatrists and psychologists, and according to the DSM-IV criteria. Also the level of depression was diagnosed by Beck test. The recording was carried out with a 19-channel device with a 10-20 standard and for 4 minutes and a sampling frequency of 256 Hz.

We first pre-processed the data and then extracted the features. To eliminate noise, the Band-pass IIR filter (10th-order Butterworth filter) was applied, with frequency scale ranging from 0.5 to 40 Hz. Then the remaining noise was detected and removed, but remained non-noisy sections, did not reassembled. After that, the remaining pieces were divided into 8-second non-overlapping windows to prevent over-fitting.

B. Feature Extraction and Dimention Reduction

The features used in this study were selected according to previous studies in this area. Features were extracted separately from each channel and were given to the system in order. Features include: the statistical features (mean, variance, skewness, kurtosis), the frequency features (Delta, theta, alpha, beta and gamma frequency sub-bands), coefficients of 10^{th} order of the AR model, RQA, nonlinear features (Higuchi fractal dimension, Katz, Correlation dimension, Lempel-Ziv complexity criterion). To prevent over-fitting we used dimension reduction (DR) methods such as t-SNE and KPCA (Gaussian kernel). After that, all features have been normalized.

C. Evaluation

To evaluate the results, two methods were used including: percent relative difference(PRD) and Pearson correlation. PRD represents normalized sum of the squared error as the percentage, and the Pearson correlation shows how much the estimated outputs track the actual values. Below are corresponding equations:

$$PRD\% = \sqrt{\frac{\sum_{i=1}^{N}(x_{original}(i) - x_{model}(i))^2}{\sum_{i=1}^{N}(x_{original}(i))^2}} \times 100\% \qquad (1)$$

$$corr(X,Y) = \frac{\mathrm{cov}(X,Y)}{\sigma_X \sigma_Y} \qquad (2)$$

D. Quantification Algorithm

In this research, we present a novel approach for indexing of depression. In fact we want to rebuild the BDI. The block diagram is shown in Figure 1. In this approach, the numeric index is obtained by using the classifiers that assigns the membership values to each of the classes. For this purpose, we need classifiers that generate the membership values of the classes. In this study, two classifiers were used, including: fuzzy Ishibuchi classifier[10] and SVM. In fuzzy classifier the membership values are at hand, but in SVM method, using the solution provided by the Platt[11], the outputs of the standard SVM were mapped to probability of belonging to classes. In this method, probabilities are calculated based on the distance of test samples from the hyper-plane. In the standard linear SVM, we used following equation to obtain the score that classifier assigns to samples for classification:

$$f(x) = w^T x + w_0 \qquad (3)$$

With these scores, we can obtain the probability of belonging to a class based on the Platt method where a sigmoid function is fit to SVM scores. It is defined as follows:

$$p(c_i \mid f) = \frac{1}{1 + \exp(Af(x) + B)} \qquad (4)$$

where P is the probability of belonging to class c_i and $f(x)$ is output of the standard SVM. Parameters A and B are obtained using maximum likelihood estimation from the training set.

We used the following algorithm separately for each classifier, and then we compared the results of them. The algorithm of this method is as follows:

1- Assigning a four-class classifier to each Channel and train the corresponding data on them.

2- Training of membership values of learning data, to MARS regression model[12]: MARS model makes flexible models by fitting piecewise linear regressions; the non-linearity of a model is approximated through the use of separate linear regression slopes in different intervals of the independent variable space. Therefore the slope of the regression line is allowed to change from one interval to the other as the two knot points are crossed. The variables to be used and the end points of the intervals for each variable are found through a fast but intensive search procedure. In addition to searching variables one by one, MARS also searches for interactions between variables, allowing any degree of interaction to be considered as long as the built model can better fit the data[13]. The general MARS function can be defined as the following equation:

$$f(x) = a_0 + \sum_{m=1}^{M} a_m \prod_{k=1}^{K_m} [s_{km}(x_{v(k,m)} - t_{km})]_+ \qquad (5)$$

where a_0 and a_m are parameters, M is the number of basis functions, K_m is the number of knots, $s_{km}=\{1,-1\}$ and indicates the right/left sense of the associated step

Figure 1: Block diagram of the proposed methodology for EEG-based estimation of depression index

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function, $v(k,m)$ is the label of the independent variable, and t_{km} indicates the knot location.

3- Obtaining the membership values of the test data: It is obtained by the 19 classifiers which had been learned in step 1. Then, we have 19 quaternion(four membership values for every 19 channels) for each window, in hand.

4- Calculating the weight of each window: To calculate the numerical index of a subject, first, numerical indices of each window of the subject are estimated and then a weight is assigned to each window:

$$\beta_{C_T} = \sum_{\mu_i \in C_T} \mu_i \qquad C_T = \{C_1, C_2, C_3, C_4\} \tag{6}$$

where C_T is the class's number and μ_i is membership value of the winning class(the class with maximum membership value) for each channel separately. In the following, the weight of windows is calculated:

$$\beta_{Cx} = \max\{\beta_{C_1}, ..., \beta_{C_4}\} \tag{7}$$

$$\beta = \sum_{T=1, T\neq T}^{c} \beta_{C_T} / (c-1) \tag{8}$$

$$m_k = \{\beta_{C_x} - \beta\} \Big/ \sum_{T=1}^{c} \beta_{C_T} \tag{9}$$

where c is the total number of classes which is equal to 4 and m_k is the calculated weight for the k^{th} window of the subject. The above method, in fact, considers a higher weight for the windows announcing the winner class with the most votes, thus gives a more accurate estimate of the subject's numerical index.

5- Combining the membership values of channels by the Dempster-Shafer method[14]: For this purpose, the 19 quaternion, are combined into one quaternion.

To do this, we must first determine the Basic Probability Assignments (BPA) that also defined based on the probabilities of belonging to classes which computed from two classification methods previously. Dempster-Shafer allows for representation of both imprecision and uncertainty through the definition of two functions: plausibility *(Pls)* and belief *(Bel),* both derived from a mass function m (or basic probability assignment). These BPAs shall satisfy following two conditions:

$$m : P(D) \rightarrow [0,1] \tag{10}$$

$$m(\phi) = 0, \sum_{A \in P(X)} m(A) = 1 \tag{11}$$

where m is mass function(or bpa)[15, 16]. The set $P(D)$ is a power set on D, and A is a subset on this set. Mass functions are defined on the power set of the space of discernment D, i.e. a mass is attributed to each subset of D. In classification problems, D may for instance be the set of classes of interest, and a subset of D represents a union of classes[17]. Belief and plausibility functions are derived from the mass function, and are respectively defined by:

$$Bel(\varnothing) = 0, \quad Bel(A) = \sum_{B \subset A} m(B), \forall A \subset D, A \neq \varnothing \tag{12}$$

$$Pls(\varnothing) = 0, \quad Pls(A) = \sum_{B \cap A \neq \varnothing} m(B), \forall A \subset D, A \neq \varnothing \tag{13}$$

Now, if we have evidence issued from several sources, which are modelized in the DS framework by means of the previously defined functions, these masses are combined by the orthogonal rule of Dempster[16]. For m_j being the mass function associated with source j ($j = 1, \ldots, n$), this rule is written, for all non-empty subset A of D:

$$(m_1 \oplus m_2 \oplus ... \oplus m_n)(A) = \frac{\sum_{B_1 \cap ... \cap B_n = A} m_1(B_1) m_2(B_2) ... m_n(B_n)}{1 - \sum_{B_1 \cap ... \cap B_n = \varnothing} m_1(B_1) m_2(B_2) ... m_n(B_n)} \tag{14}$$

$$(m_1 \oplus m_2 \oplus ... \oplus m_n)(\varnothing) = 0 \tag{15}$$

If this expression is defined, i.e. if:

$$k = \sum_{B_1 \cap ... \cap B_n = \varnothing} m_1(B_1) m_2(B_2) ... m_n(B_n) < 1 \tag{16}$$

Similar equations can be derived for directly combining belief or plausibility functions. To some extent, k can be interpreted as a measure of conflict between the sources and is directly taken into account in the combination as a normalization factor.

6- Calculating the numerical index of each window of the subject(I_k), by the MARS model trained in step 2: For this purpose, we feed the quaternion derived from the Dempster-Shafer combination to MARS model, and consider the output as the numerical index of the corresponding window of the subject.

7- Calculating the subject's depression index: It is derived with the weighted average:

$$Index_P = \frac{\sum_{k=1}^{W} m_k I_k}{\sum_{k=1}^{W} m_k} \tag{17}$$

where W is the total number of windows of the P^{th} subject and I_k is the estimated index value of the k^{th} window of the same subject.

III. Results and Discussion

The results are obtained for all categories of features in addition to various dimension reduction methods. Five types of features were investigated in this study. As shown in Figure 2, the best results are obtained from the RQA and nonlinear features. Moreover, the fuzzy classifier shows less 3 to 4 percent PRD compared to SVM which indicates our data is better discriminated with fuzzy classifier. Nonlinear features, on the other hand, due to the nonlinear and complex nature of the EEG signal, showed a higher ability to detect the latent information in the signal. Since the best result was obtained from the RQA features, we used this type of feature in other experiments.

In Figure 3, the dimension reduction algorithms are compared. As seen for all models, the tSNE method has had the best performance. The reason for these results is the way in which tSNE performs all the features to maintain information while retaining the separability. However, the KPCA method has less execution time than tSNE, which is a significant advantage. In addition, the genetic algorithm was used to find the optimal parameters of Gaussian kernel of KPCA method. In this study, the optimum dimension is also selected equal to 5 using trial and error.

As explained in the previous section, the Dempster-Shafer method was used to combine the results of the channels, but to realize the effect of this combination method, we considered a state that used a simple averaging instead of dempster-shaffer. In Figure 4, the use of the Dempster-Shafer method has improved the performance of the system. In fact, this method, due to the integration of different channel information, results in a more accurate estimate of the Beck index.

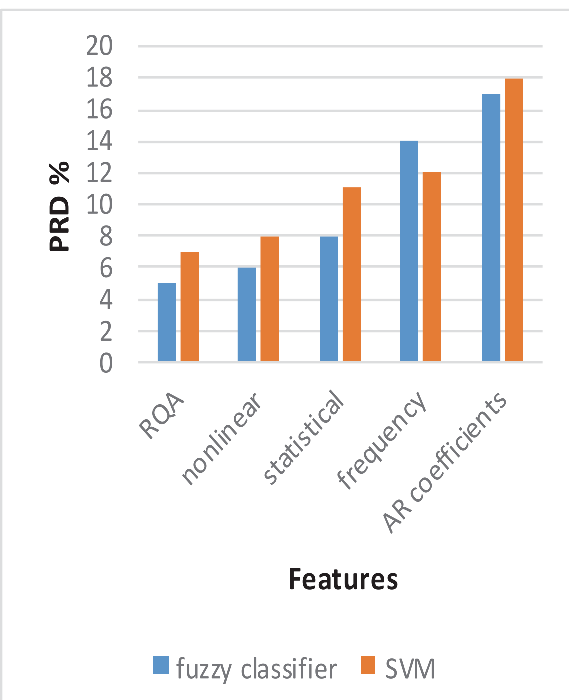

Figure 2: Comparison of different attributes with t-SNE method

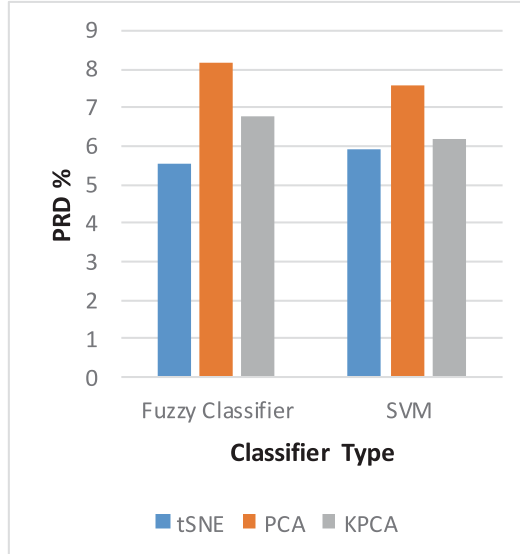

Figure 3: Comparison of Different DR Methods for RQA features

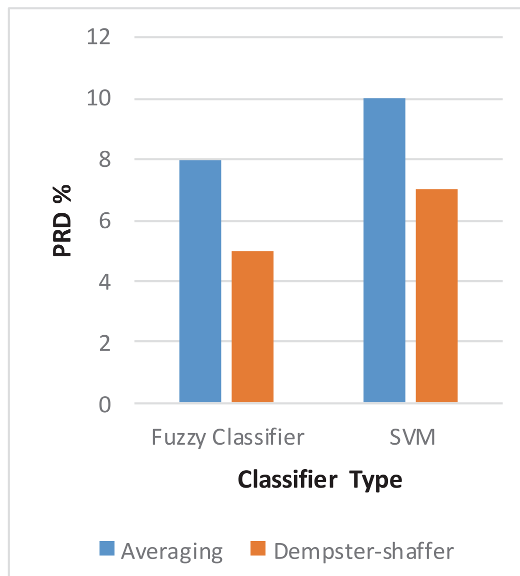

Figure 4: Comparison of Dempester-Shafer method with tSNE and RQA features

IV. Conclusion

In this study, a novel algorithm proposed to estimate a numerical index of the severity of depression. For this purpose, different linear and nonlinear methods employed to extract the features from EEG signals. Moreover, dimension reduction methods were utilized to preserve the generalization ability of classifier. A total number of 75 subjects were used in

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this research, and BDI of every subjects was at hand, and they were categorized to 4 levels of depression. Since there are 19 channels for each subject, the Dempster-Shafer method was used to combine the information of different channels. Further research is being done in this area to reduce the risk of depression diagnosis.

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